

1.4 HW Gavin Anderson

4. Let n be an integer. Use Definition 1.6 to explain why $2n + 7$ is an odd integer.

1.6) an integer n is odd if $n = 2k + 1$ for some integer k .

$$\text{Proof: } 2n + 7 = 2n + 6 + 1 = 2(n+3) + 1$$

Since n is an integer, $n+3$ is too. $k = n+3$. Then $2n + 7 = 2k + 1$ for this integer k .

6. Consider the following definition of the " $\triangleleft$ " symbol.

Definition. Let x and y be integers. Write $x \triangleleft y$ if $3x + 5y = 7k$ for some integer k .

- (a) Show that $1 \triangleleft 5$, $3 \triangleleft 1$, and $0 \triangleleft 7$.

- (b) Find a counterexample to the following statement:

If $a \triangleleft b$ and $c \triangleleft d$, then $a \cdot c \triangleleft b \cdot d$.

a) using the def.

$$1 \triangleleft 5: 3(1) + 5(5) = 28 = 7(4)$$

$$3 \triangleleft 1: 3(3) + 5(1) = 14 = 7(2)$$

$$0 \triangleleft 7: 3(0) + 5(7) = 35 = 7(5)$$

$$\text{b.) } a=1, b=5, c=3, d=1$$

$$a \triangleleft b \quad \& \quad c \triangleleft d \quad \text{both hold}$$

$$(1 \ 5) \quad (3 \ 1)$$

$$\text{But } ac=3 \quad \& \quad bd=5 \quad \& \quad 3(3) + 5(5) = 9 + 25 = 34 \quad (\text{not a multiple of } 7)$$

$$7(4) + 6 = 34$$

$$a \triangleleft b \quad \& \quad c \triangleleft d \quad \text{hold}$$

$$\text{but } a \cdot c \triangleleft b \cdot d \quad \text{fail}$$

8. There are several different models for geometries in which the points are ordered pairs (x, y) of real numbers; we plot these points in the usual way in the xy -plane. In such a geometry, there can be a formula for the distance between two points (x_1, y_1) and (x_2, y_2) . For example, in Euclidean geometry, distance is given by the usual Euclidean distance formula:

$$\text{Distance} = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

In any geometry with a distance formula, we can define a *circle* as follows.

Definition 1.9 A circle centered at (a, b) with radius r is the collection of all points (x, y) whose distance from (a, b) is r .

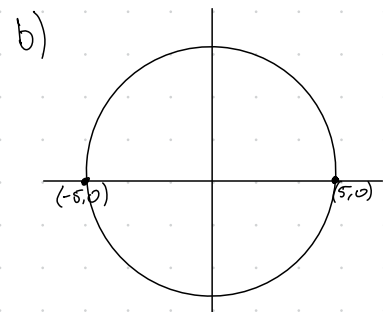
- (a) Use Definition 1.9 to give an equation for the circle with radius 5 centered at $(0, 0)$ in the Euclidean plane.
 (b) Plot the circle from part (a) in the xy -plane.
 (c) In the *Taxicab geometry*, the distance between two points (x_1, y_1) and (x_2, y_2) is given by the following formula.

$$\text{Distance} = |x_2 - x_1| + |y_2 - y_1|$$

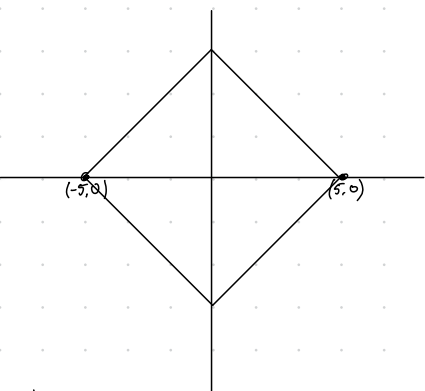
(This is called "taxicab" distance because it models the distance you would have to travel if you were restricted to driving on a rectangular city grid.) In this model, use Definition 1.9 to plot the "circle" with radius 5 centered at $(0, 0)$.

- (d) Which type of circle (Euclidean or taxicab) agrees with your concept image of circle?

a) $(a,b) = (0,0) \notin r=5$
 $\sqrt{x^2 + y^2} = 5 \rightarrow x^2 + y^2 = 25$



c) $|x| + |y| = 5$



d) Euclidean, its normal to me, but it depends on how you measure the distance.

14. Let $P(n, x, y, z)$ be the predicate " $x^n + y^n = z^n$."

(a) Write the following statement in predicate logic, using positive integers as the domain.

For every positive integer n , there exist positive integers x, y , and z such that $x^n + y^n = z^n$.

(b) Formally negate your predicate logic statement from part (a). Simplify so that no quantifier lies within the scope of a negation.

(c) In order to produce a counterexample to the statement in part (a), what, specifically, would you have to find?

a) $(\forall n)(\exists x)(\exists y)(\exists z)P(n, x, y, z)$

b) negation & Push

$(\exists n)(\forall x)(\forall y)(\forall z)\neg P(n, x, y, z)$

"There exists a positive integer n such that for all positive integers x, y, z , $x^n + y^n \neq z^n$."

c) Show negation in (b) is true. You would have to find one specific value of n for which no positive integers x, y, z at all satisfy $x^n + y^n = z^n$.

16. Find a counterexample for each statement.

- (a) If all the sides of a quadrilateral have equal lengths, then the diagonals of the quadrilateral have equal lengths.
- (b) For every real number $N > 0$, there is some real number x such that $Nx > x$.
- (c) Let l, m , and n be lines in the plane. If $l \perp m$ and n intersects l , then n intersects m .
- (d) If p is prime, then $p^2 + 4$ is prime.

a) A Rhombus other than a Square. All four sides are equal length by def. of a Rhombus, but its diagonals have different lengths.

b) Take $N=1$. Then $Nx = x$ for every real x , so $Nx > x$ becomes $x > x$ which is false no matter what x is.

c) $l = \text{vertical line}$

$m = \text{horizontal line (so } l \perp m)$, $\exists$ a different horizontal line, parallel to m . Then:

n intersects l

n does not intersect m

Satisfies hypothesis, breaks conclusion

d) Take $p=2$ (Prime). Then $p^2 + 4 = 8$, which is not prime.

18. Consider the following theorem.

Theorem. Let x be a wamel. If x has been schlumped, then x is a borfin.

Answer the following questions.

- (a) Give the converse of this theorem.
- (b) Give the contrapositive of this theorem.
- (c) Which statement, (a) or (b), is logically equivalent to the Theorem?

a) Converse Swap $P \leftrightarrow Q$

Let x be a wamel. If x is not borfin, then x has not been schlumped.

b) Contrapositive Negate & Swap

Let x be a wamel. If x is not a borfin, then x has not been schlumped

c) The Contrapositive, (b), is logically equivalent to the original theorem. Conditional statement = Contrapos. always have the same truth value; Converse is a different statement that can be true or false.